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PATENT INFORMATION EXTRACTION FORM

TOPIC : Autonomous Neuromorphic Underwater Coral Restoration System with Event-Based Detection and Adaptive Indexed Pod Deployment

SCHOOL: Chitkara University Institute of Engineering and Technology

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Answer the following questions in brief: (use extra sheets if needed, submit pics/videos etc. which can help us to understand the invention better)

1. What was the problem?

- Current coral monitoring systems are not fully automatic, as they depend on humans to analyze data and take action later for coral damage detection and farther deployment .

- There is a delay between detecting coral damage and performing restoration, which reduces effectiveness and increases cost.

- Existing systems use traditional cameras and cloud processing, which require more energy and are not suitable for real-time response.

- There is no single system that can both detect coral damage instantly and perform restoration at the same location without human involvement

2. How did you solve it (Inventive Step)?

- We designed an autonomous underwater system that can both detect coral damage and respond without human involvement using the help of SNN algorithm.

- The system uses an event based sensor instead of a regular camera, which helps in faster detection and lower power usage.

- It processes the data directly on the device, so there is no need for cloud-based analysis.

- When damage is detected, the system automatically deploys coral restoration pods at the exact location.

- A controlled mechanism ensures accurate placement and avoids repeating the process in the same area, making it efficient and scalable.

3. What were the other possible solutions and why they could not be done?

- Several alternative solutions were consider for coral monitoring, damage detection and restoration.

- One option was using normal cameras with AI models and sending data to a server but it needs high bandwidth, more power with the help of CNN based algorithm and is not suitable for long underwater use.

- Manual inspection by divers was also considered but it is slow, expensive and not practical for covering large coral areas.

- Some basic under water robots can place coral at fixed intervals but they cannot detect damage properly and may waste resources.

- Hydraulic systems for deployment were another option but they are complex, heavy and have risks like leakage and high maintenance.

- Because of these issues we chose a smarter system that can detect damage and restore coral in real time in an efficient and automatic way.

4. What are the advantages of the solution proposed by you?

- **Real-Time Autonomous Operation** : The system can detect coral damage and immediately take action at the same location without waiting for human input or external analysis.

- **Energy Efficiency** : It uses an event-based sensor and efficient processing, which helps reduce power consumption compared to traditional camera systems using the help of SNN algorithm.

- **Reduced Delay** : Since all processing happens on the device itself, there is no need to send data to servers, which makes the response much faster.

- **Adaptive Deployment Mechanism**: The system can place restoration pods accurately based on how severe the damage is using x-y electric gantry (printer works mechanism), ensuring better results.

- **Scalability** : It can work continuously over large coral reef areas without needing divers, making it suitable for large-scale use.

- **Improved Accuracy** : The system remembers previous locations and avoids placing pods again in the same spot, reducing waste.

- **Mechanical Reliability** : The electric mechanism is simpler more reliable,and requires less maintenance compared to complex hydraulic systems.

- **Bandwidth Optimization** : It processes only important changes instead of full images which leads to reduces data usage and communication needs.

- **<u>Integrated System :</u>** Unlike the conventional systems used before that separate monitoring and restoration the invention combines both into a unified autonomous platform.

5. Explain the step wise working of the innovation explaining all the components used in the invention and the specific function they are performing.

- **<u>Step 1</u>: Underwater deployment and mobility :** The device is deployed underwater and navigates to the targeted reef areas using the thruster based propulsion.

Components used:

- Waterproof pressure sealed enclosure
- Thrusters (vectored fans) and ESC drivers
- IMU (orientation/tilt)
- Depth and pressure sensor
- Battery pack and power management module

Function: It provides stable underwater movement, depth holding and electrical power distribution.

- **<u>Step 2</u>: Real time reef scanning using event driven vision :** The device continuously scans coral surfaces using an event driven camera. Only pixel level intensity changes are captured thus reducing data load.

Components used:

- Event driven camera module
- Dual LED illumination (fixed)
- Optical lens and mounting frame

Function: It detects coral texture change, loss of living tissue, fragmentation and low biological activity with low power and low latency.

➢ **Step 3: Coral spawning or egg sensing mode :** When the system enters a spawning window or detects a local particle increase, it activates the spawning and egg detection module.

Components used (possible solution):

- Macro optical sensor or micro camera and LED
- Turbidity or particle sensor (flow through)
- Micro pump and micro filter sampling chamber

Function: It detects coral gamete bundles and larvae presence through optical classification and particle change confirmation, and logs spawning locations for research and future restoration planning.

➢ **Step 4: Neuromorphic processing and coral health classification :**

The event stream and optional egg sensor data is processed by locally using a Spiking Neural Network (SNN) running on the onboard processor.

Components used:

- SNN or neuromorphic processing unit AI chip
- Embedded MCU ESP32 or STM32 for control
- Local storage (Flash or SSD)

Function: It classifies coral health as **Healthy, Damaged or Decomposed** and generates a confidence score for decision making.

➢ **Step 5: Decision logic and safety checks:** When the coral is classified as decomposed or damaged the microcontroller performs control checks before deploying the pods.

Decision conditions include:

- Confidence score above threshold
- Stable hovering (IMU and depth confirmation)
- No obstacle and jam detection
- Treated location verification (memory)

Components used:

- MCU firmware logic
- Jam detection sensor (gate/gantry feedback)
- Localisation/mapping module

Function: It prevents false deployment and duplicate deployment at the same spot, and ensures safe operation.

➢ **Step 6: XY electric gantry indexing to select pods (10×10 grid) :**

After confirmation, the indexed electric gantry positions itself to the selected pod coordinates in the 10×10 storage grid.

Components used:

- X axis stepper motor and linear rail
- Y axis stepper motor and linear rail
- Position encoder/limit switches
- Pod storage grid (100 box pods)

Function: It provides accurate positioning for controlled pod selection and avoids mechanical misalignment.

➢ **Step 7: Automatic multi pod release (10 pods) :** Once aligned, the system releases **exactly 10 pods** through the dispensing gate.

Components used:

- Servo actuated release gate
- Pod counter sensor (IR/limit/hall)
- Discharge chute/outlet port

Function: Ensures precise quantity based deployment for coral regeneration while avoiding over release.

➢ **Step 8: Logging, mapping, and communication :** The device stores the event data, coral health status, pod count released, and the treated location. If communication is available it transmits summary data to a surface/server.

Components used:

- Storage module (SSD/Flash)
- Communication module (acoustic modem / RF at surface)
- GPS at surface and underwater localisation (as available)

Function: It maintains restoration records, enables revisit planning, and supports long term reef monitoring.

➢ **Step 9: Continue scanning :** The device resumes scanning and repeats the above steps for new damaged coral regions.

6. Attach drawing hand made/ computer made showing all the components of the invention.

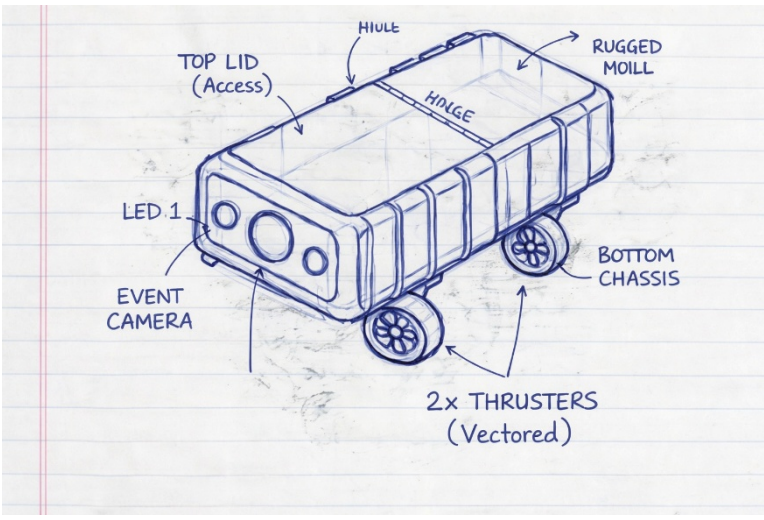

Figure 1: A rugged coral restoration device featuring a top access hinged lid, front mounted event driven camera with dual LEDs, and vectored thrusters for stable navigation and deployment.

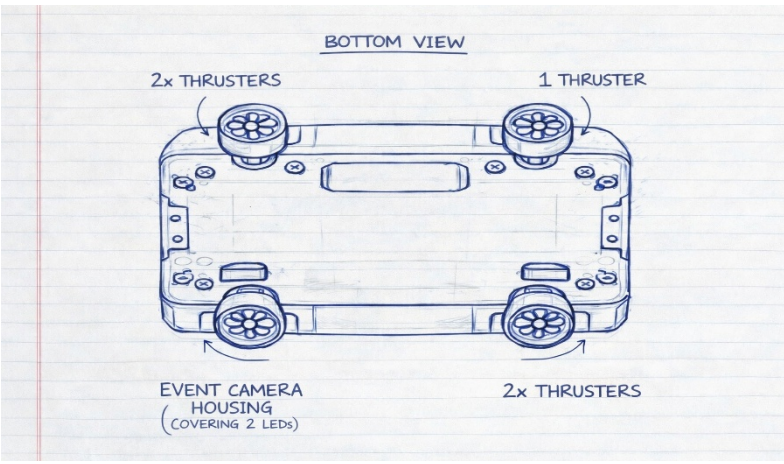

Figure 2: Bottom view showing four vectored thrusters and a protected event camera housing with dual LED coverage for underwater stability and vision.

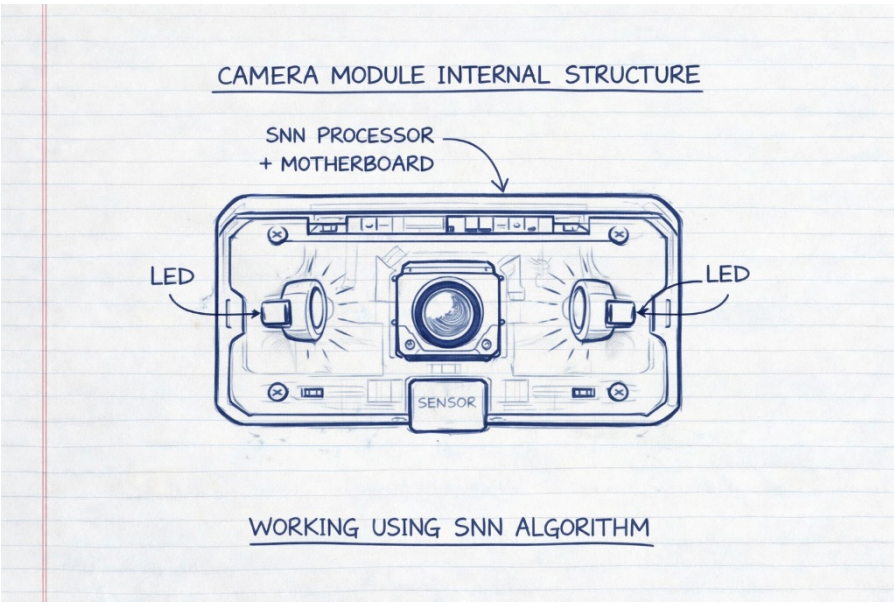

Figure 3: Internal camera module featuring a central event driven sensor with dual LED's and an integrated SNN processor for real time underwater reef analysis**.**

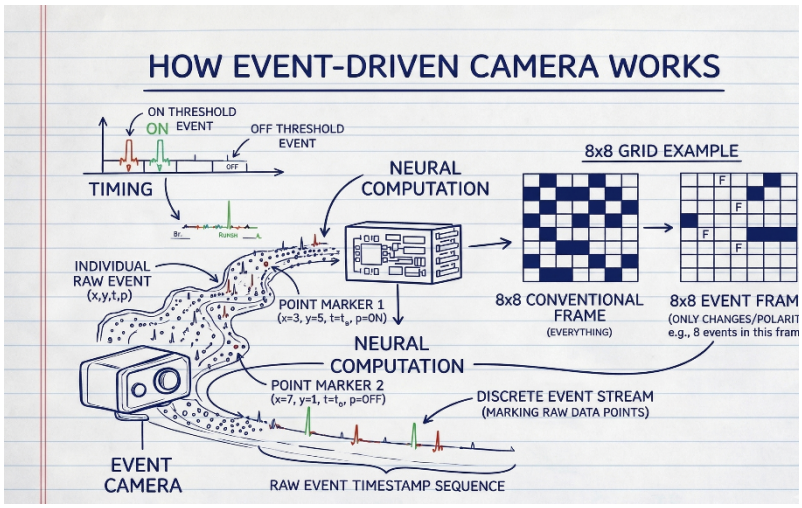

Figure 4: An **event driven camera** captures only changes in brightness at each pixel and sends them as fast, spike like signals instead of recording full image frames continuously.

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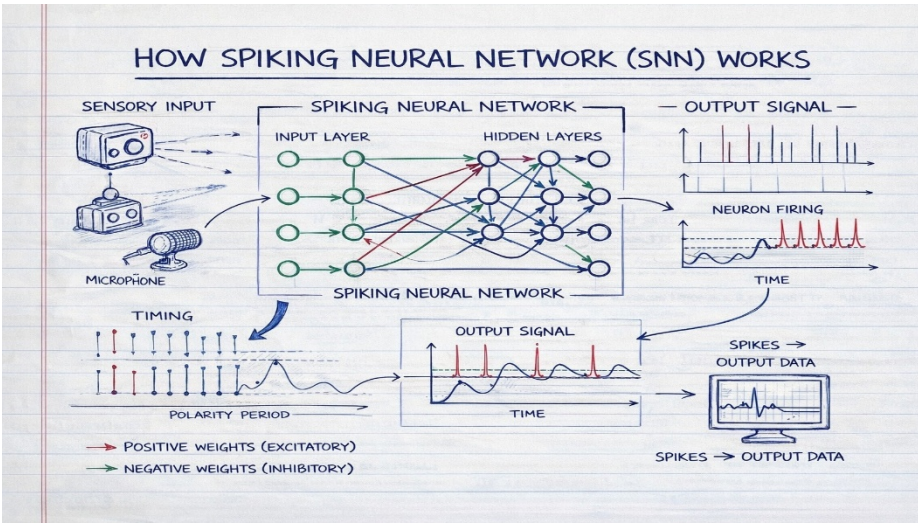

Figure 5: A Spiking Neural Network (SNN) processes time based sensory inputs as discrete spikes, using excitatory and inhibitory neuron firing patterns to generate efficient and event driven output signals**.**

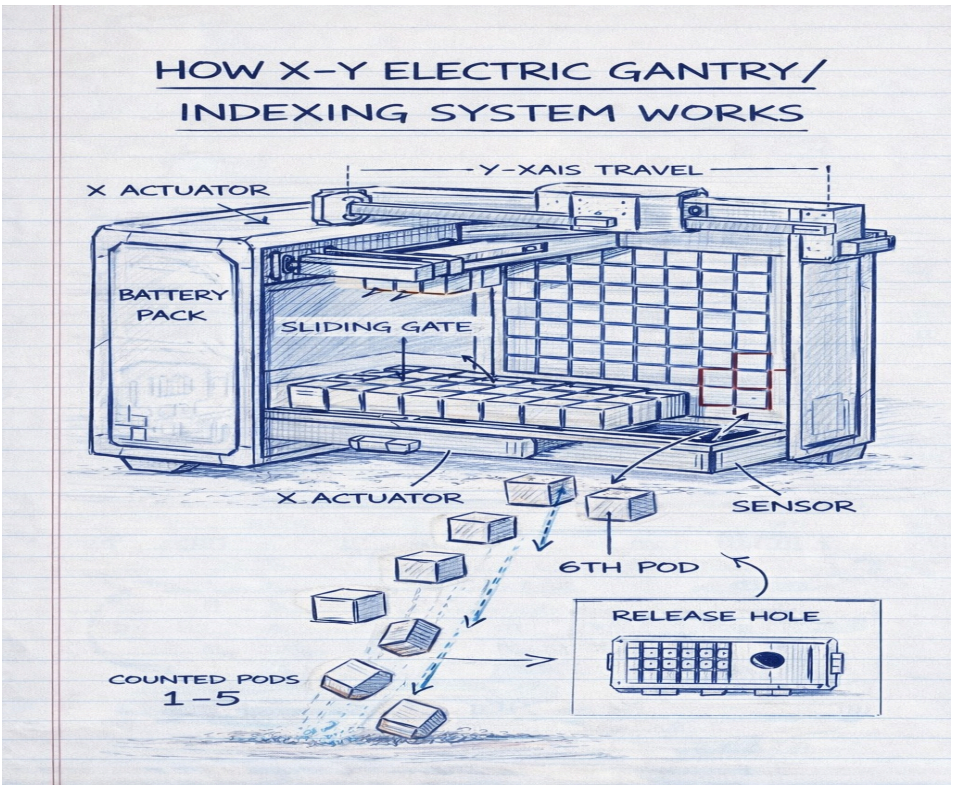

Figure 6: The XY electric gantry indexing system precisely moves across a pod grid using actuators and sensors to align and release selected coral pods through a controlled deployment hole**.**

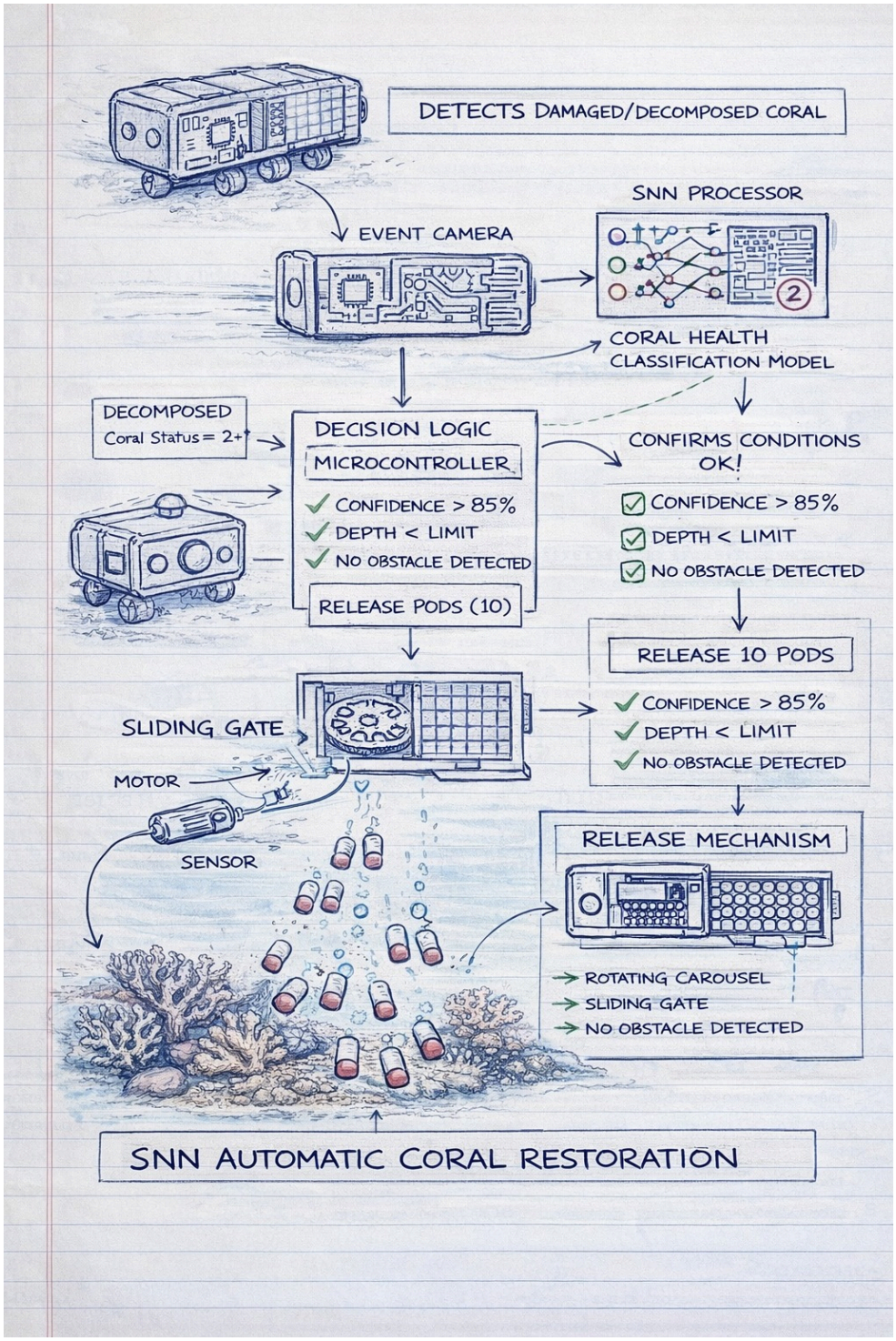

Figure 7 : An SNN based autonomous system detects damaged coral, validates environmental conditions and automatically deploys biodegradable pods through a controlled release mechanism for reef restoration.

Relevant Key phrases relating to your invention

S.NO.	Key phrases
1	Neuromorphic Event-Based Underwater Coral Detection System
2	Autonomous SNN-Based Real-Time Coral Health Classification
3	Adaptive XY Indexed Electric Gantry Pod Deployment Mechanism
4	Integrated Detection-to-Restoration Autonomous Marine Rehabilitation Platform